1. Problem and decision boundary

- Separate weak transport collision evidence from normal payload/support wrench.
- A=servo, B=torque; final controller is unchanged from v1.
- Finite-condition utility, not safety certification.

2. Evidence audit and independence

- Official evaluate.py was consumed while debugging L1: compatibility evidence only.
- Fresh development, D4 and sealed holdout support final efficacy claims.
- No post-holdout retuning; all misses, masked cases and false rises remain.

3. Original learner: method and motivation

- Normal-operation-only 77-basis ridge predicts v1 horizontal wrench from causal reference, mass and COM inputs.
- Target=w_v1,xy; train mean/scale/weights on train only; threshold once on disjoint no-collision validation.
- e=w_v1,xy-Phi beta; EMA alpha=.1; 4-cycle persistence, 8-cycle release.
- Motivation: remove motion/load baseline to expose weak horizontal force.

4. L3 is a gate bugfix, not relearning

- active=grip AND min |z_bottom(q_prev)-h_table| > unchanged band.
- Exact ORIGINAL geometry-height contract; v1 OR retained.
- Weights, threshold, EMA and persistence unchanged. L3 exact ORIGINAL geometry-height repair adopted

5. Frame, power and finite geometry audit

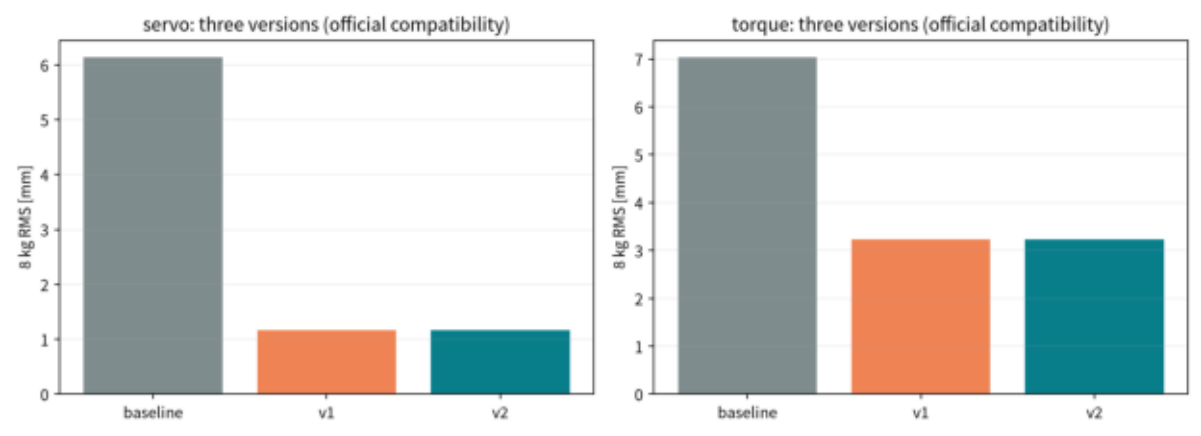
- f_q=f_p; tau_q=tau_p+(p-q)x f_p; v_p=v_q+omega x(p-q).
- Power invariant W·V and joint virtual power W·Jqdot=tau·qdot were checked.
- Finite frame/Jacobian tests catch sign errors; they do not prove global geometry, observability or safety.

6. Candidate failures

- C bounded-normal: G4 0/4 for each architecture.
- Fixed .10 s return plus 1.5 mm/.01 m/s bounds cannot cover the entire state box; not every tested trajectory is claimed to violate bounds.
- L1 official FA regression; L2 exploratory train box; L3 exact ORIGINAL geometry-height repair adopted

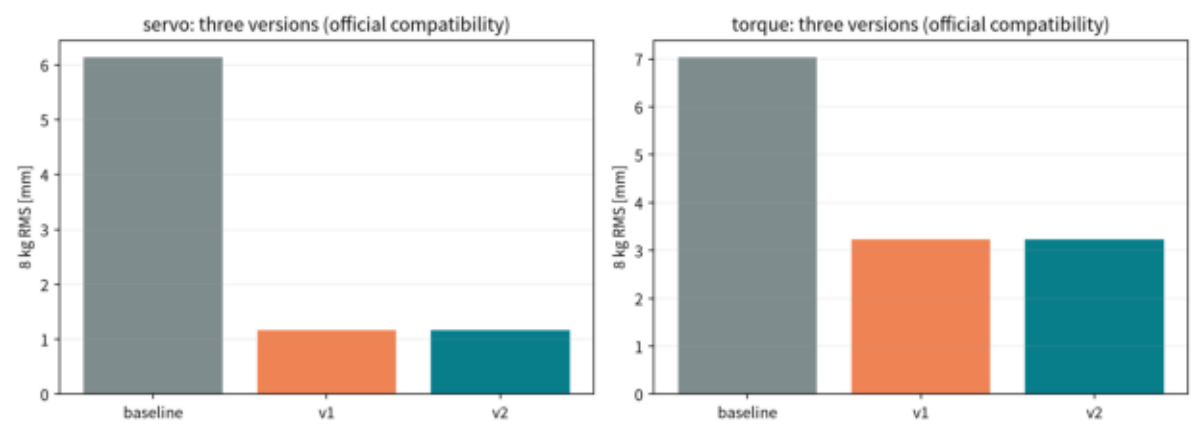
7. Official compatibility: A

- baseline: TCP 6.14/14.00 mm; step 57.5/73.8 N m; table 101 N; FA 5/2/0; transport 15 N 68 ms, 10 N 놓침, min 15 N
- v1: TCP 1.17/3.91 mm; step 9.7/14.6 N m; table 79 N; FA 0/0/0; transport 15 N 83 ms, 10 N 놓침, min 15 N
- v2: TCP 1.17/3.91 mm; step 9.7/14.6 N m; table 79 N; FA 0/0/0; transport 15 N 45 ms, 10 N 62, min 10 N
- Post-debugging cells: not independent efficacy evidence.



8. Official compatibility: B

- baseline: TCP 7.04/13.18 mm; step 57.2/74.8 N m; table 295 N; FA 5/4/0; transport 15 N 68 ms, 10 N 97, min 10 N
- v1: TCP 3.23/8.44 mm; step 9.0/13.1 N m; table 80 N; FA 0/0/0; transport 15 N 83 ms, 10 N 놓침, min 15 N
- v2: TCP 3.23/8.44 mm; step 9.0/13.1 N m; table 80 N; FA 0/0/0; transport 15 N 43 ms, 10 N 56, min 10 N
- Same unchanged evaluator and caveat.

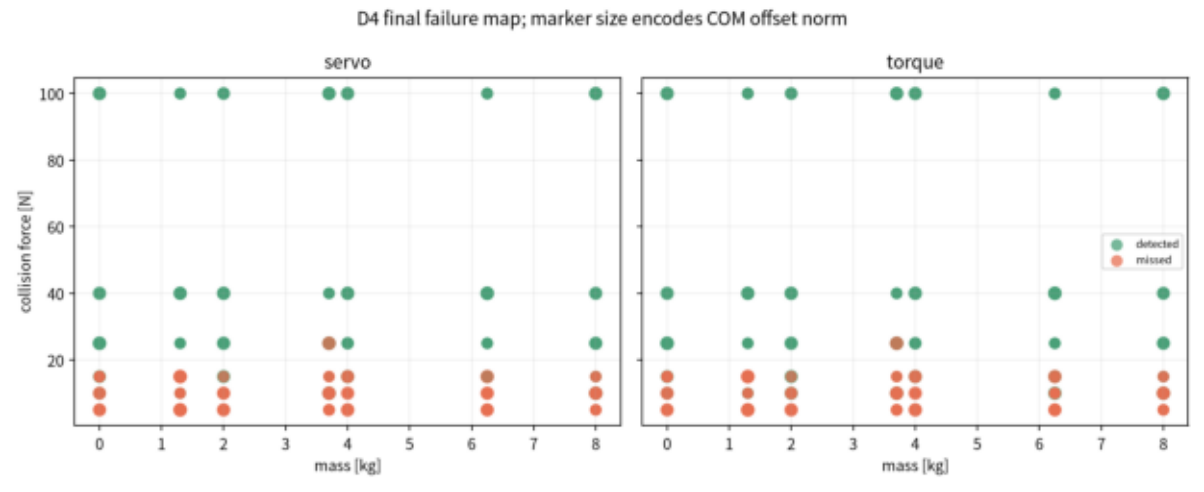


9. Fresh development: v1 versus final

- All collision classifications
- L3/servo: detected 28, missed 36, masked 0, collision-extra FA 0, pure-ref FA 0
- L3/torque: detected 38, missed 26, masked 0, collision-extra FA 0, pure-ref FA 0
- v1/servo: detected 28, missed 36, masked 0, collision-extra FA 0, pure-ref FA 0
- v1/torque: detected 27, missed 37, masked 0, collision-extra FA 0, pure-ref FA 0

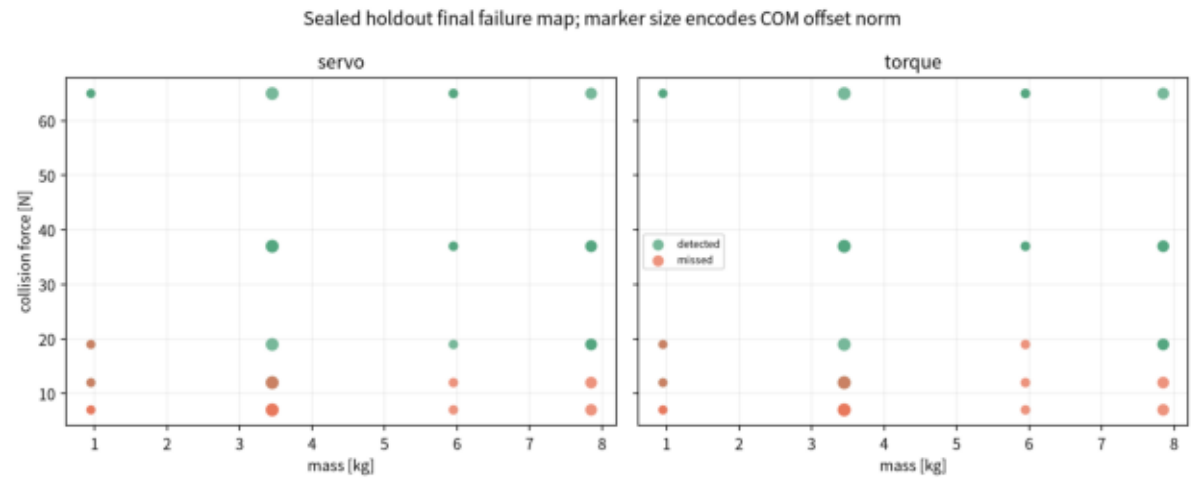
10. D4 mass/COM failure map

- 385 cases per architecture/version
- v1/servo: detected 94, missed 86, masked 0, collision-extra FA 2, pure-ref FA 0
- v1/torque: detected 91, missed 89, masked 0, collision-extra FA 1, pure-ref FA 0
- v2/servo: detected 111, missed 69, masked 0, collision-extra FA 2, pure-ref FA 0
- v2/torque: detected 111, missed 69, masked 0, collision-extra FA 3, pure-ref FA 0



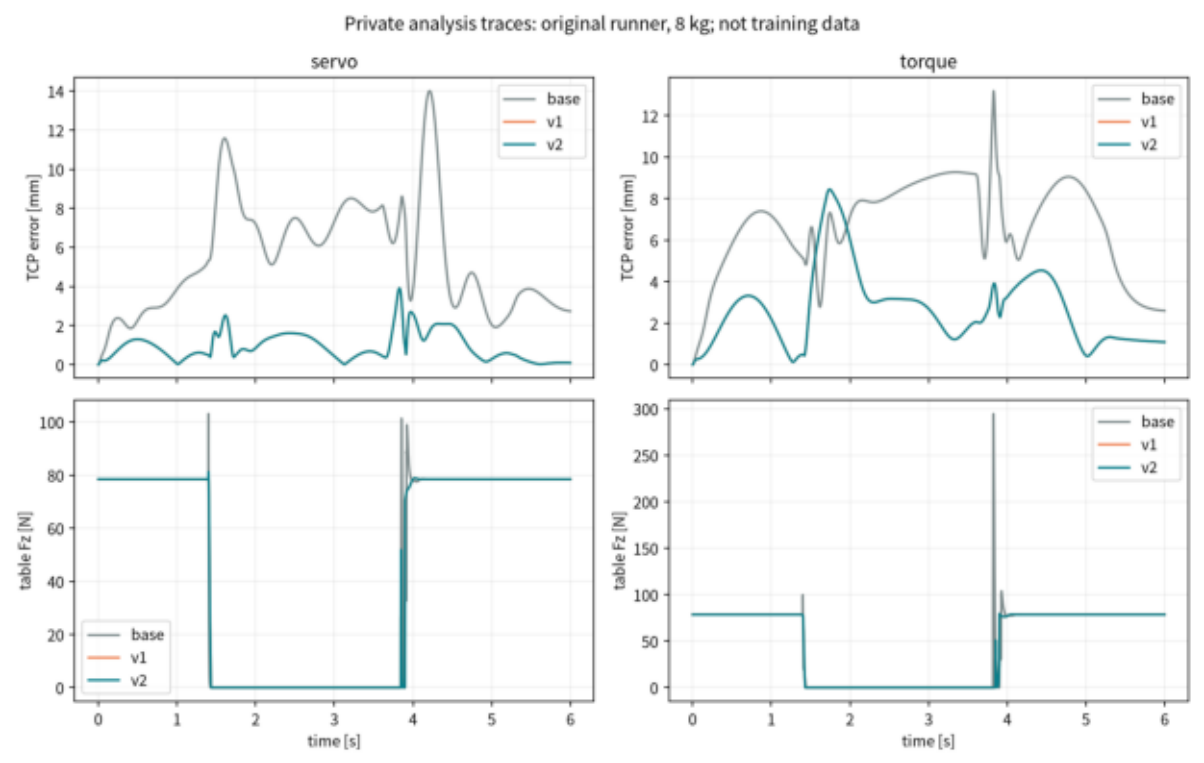
11. Sealed holdout failure map

- 38 cases per group; opened after freeze
- v1/servo: detected 15, missed 15, masked 0, collision-extra FA 0, pure-ref FA 0
- v1/torque: detected 14, missed 16, masked 0, collision-extra FA 0, pure-ref FA 0
- v2/servo: detected 19, missed 11, masked 0, collision-extra FA 0, pure-ref FA 0
- v2/torque: detected 18, missed 12, masked 0, collision-extra FA 0, pure-ref FA 0



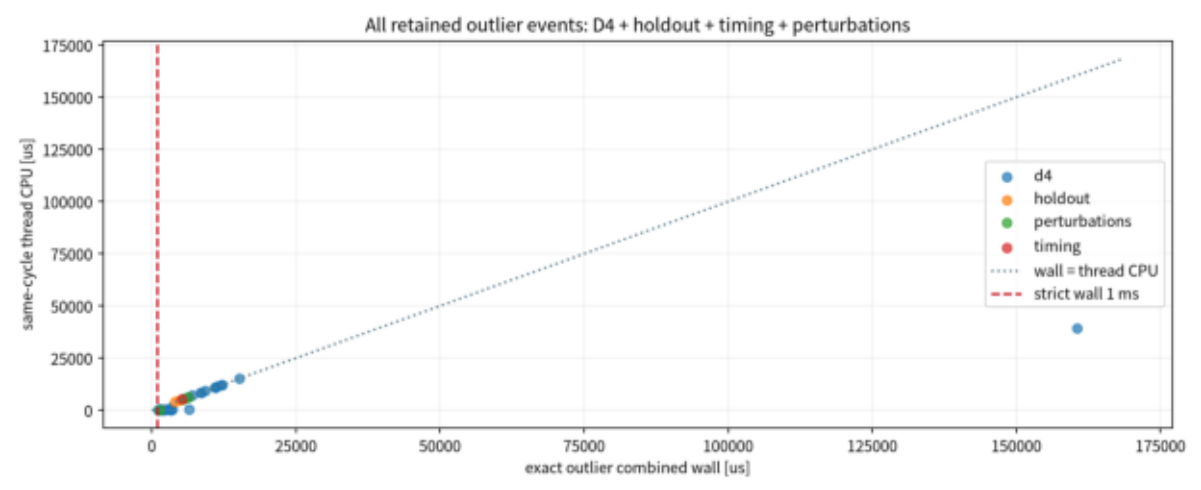
12. G4 measured support

- Final 8 kg last-20 ms support, v1 -> v2:
- servo: mean 0.002->0.002 mg; min 0.000; peak 1.004; increase 0.000; pass=False
- torque: mean 0.112->0.112 mg; min 0.004; peak 1.014; increase 0.000; pass=False
- C remains rejected; final v2 claims no controller improvement.



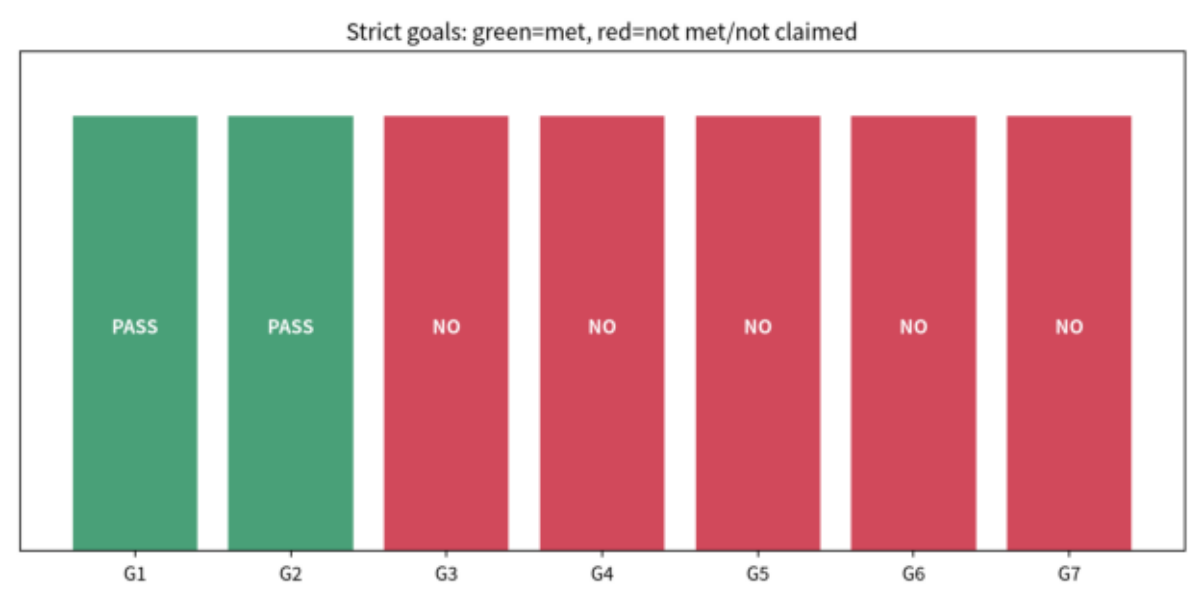
13. Timing: exact worst-cycle evidence

- v2 servo: wall 12151.7 us; exact-cycle CPU 12148.9 us; ctx v/i 0/0; pass=False
- v2 torque: wall 160578.1 us; exact-cycle CPU 39373.3 us; ctx v/i 2919/0; pass=False
- All-version comparison worst: 160578.1 us in d4; CPU 39373.3 us (thread CPU at the exact worst-wall cycle)
- G5 uses final v2 A/B only; global and v1 maxima are comparisons.



14. G1-G7 strict status

- G1: PASS
- G2: PASS
- G3: FAIL
- G4: FAIL
- G5: FAIL
- G6: FAIL
- G7: not solved; identical instantaneous vertical wrench remains ambiguous
- G1/G2 are post-debugging official compatibility observations.



15. Utility, scope and cost

- Utility: causal residual learning can expose weak horizontal force when the normal-operation baseline generalizes.
- Scope: finite simulation development/D4/sealed holdout; no probability, hardware or hard-RT claim.
- Cost: NumPy embedded weights; detector compute added, controller behavior remains v1.
- Private traces are analysis-only and never training targets.

16. Field applicability and unresolved ambiguity

- Same instantaneous feasible vertical wrench can be support or collision: G7 unresolved.
- Detector-only v2 does not improve inherited control saturation/friction/stiffness failures.
- Deployment needs sensing/process constraints, collision response and hardware validation.

